Shun Liu

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EDUCATION

SHANGHAI UNIVERSITY OF FINANCE AND ECONOMICS *(211)*
Bachelor of Computer Science *September 2021 – May 2025*

- **Selected Courses**: Algorithm Design and Analysis, Machine Learning, Database Theory, Artificial Intelligence, Python Programming, Social Network Analysis, Data Structure, Software Engineering
- **Mathematics**: Linear Algebra, Probability Theory, Statistics, Advanced Mathematic, Discrete Mathematics, Graph Theory, Operations Research
- **Research Interests**: NLP, AI4Science and Multi-Model Learning.

RESEARCH EXPERIENCE

Aug 2023 – Dec 2023 | Protein Sequence Modeling and Function Prediction [Competition]
Project Leader **Kaggle Gold Medal (Top 15/1625)**

- **Biomolecular Structure Modeling**. Based on the theory of **Gene Ontology** (GO), three directed acyclic graphs (DAG) were used to represent the biological processes, cellular components and chemical functions of protein molecules, association indicates the existence of semantic association (partial attribution, significant influence).
- **Protein Function Prediction**. The function was defined as a subset of the DAG graphs, and ProFun, QuickGO, SPROF 's proof codes were fused for GO graph reconstruction. The GO-based-method was highly expressive and efficient in downstream tasks compared to sequence alignment.

Oct 2023 – Jan 2024 | Value System and Potential Group-Dependent Bias in LLM
Research Assistant, Technical Director **Paper (Collaboration: Dr. Weicheng Ma)**

- **LLM bias.** Nowadays, the popularity of large language models (LLMs) in varied fields delivers the **LLM safety** popularity. The study of large-scale bias is in line with the cutting-edge researches of 5 trustworthy attributes: 'Fairness/Reliability/Robustness/Privacy/Interpretability', to explore whether LLM's attitudes toward different groups are consistent and influenced by human values.
- **Qualitative Analysis and Reasoning**. Qualitative experiments on Chinese-based and English-based LLM were conducted to identify and test theoretical assumptions (do LLM has certain **group-dependent value**?), which provides experimental support for the further research.

Oct 2023 -- Present | Towards Tuning PLMs for Compositional Zero-Shot Learning
Core Member **Paper (Collaboration: Dr. Zhuo Chen)**

- **Research Directions**. Based on the pre-training language model, the ability of visual-text comparison is transferred to solve the problem of multi-model alignment, the low-resource learning (CZSL) is adopted to alleviate the high cost of sample annotation and cross-modal alignment.
- **Proposed Methodology**. By leveraging the prowess of **pretrained language models** (PLMs), attribute attention weights are proposed to capture the multi-granularity information on instances.

July 2023 – Oct 2023 | Interpretable Machine Learning in High-Dimensional Sports Statistics [Paper]
Research Assistant **Paper (Mentor: Dr. Jiaxi Yang)**

- **Data Wrangling.** PCA is employed to extract top10 most representative features, hence to characterize the offensive and defensive pictures (patterns) in player-level and team-level.

- **Model Evaluation**. A comprehensive data analysis pipeline is designed, 4 ML models are applied to predict player role(the prediction accuracy is **80%**) and **#team_win** in inference stage.
- **Interpretability**. For the regression model, we can visualize the feature weight graph and the importance of the feature with the SHAP value, and for the classification task such as decision tree, we can visualize the decision path on the tree to alleviate the agnosticism of the black box model.

INDUSTRY EXPERIENCE

Aug 2023 – Jan 2024 | Artificial Intelligence Research Institute, ZheJiang Lab
Research Assistant **Mentor:** Dr. Hongsheng Wang | Prof. hengyu Zhang

- **Research Topics.** Tracking the latest research in the field of 3D human body reconstruction in computer vision, exploring how to build a human joint rotation distribution model, and focusing on the reconstruction of spatial-temporal information, by flow-based or diffusion-based methods.
- **Academic Writing**. Participate in conference documentation and patent writing (second leader in charge) , focus on multi-modal text-driven digital avatar generation, this research leverages the alignment of cross-modal data (text-image) and the training strategy for learning from few samples.

Nov 2023 – Jan 2024 | China National Nuclear Corporation (CNNC) 8th Research Institute
CV Algorithms Intern

- **Industrial CV algorithms engineering**. A real-time and efficient computer vision algorithm is designed for industrial defect detection. The detection recall rate is higher than 99%.
- **Web development**. Develop web application of multi-instrument real-time monitoring penal and digital assistant Q&A user interface using unified *vue*/*uniapp*/*flask* framework.

Jul 2023 – Nov 2023 | Research Institute of Western Securities CO., LTD
Analyst Assistant

- **AI product Fundamental analysis**. In-depth independent analysis of AIGC/LLM/autonomous driving industry, pay attention to industry cycle, volatility, competition pattern and market size, master mainstream large model application scenarios, parameters, etc.
- **LLM Tracking & Evaluation**. To test the language models at home and abroad (GPT-3.5/GPT-4/Wen Xin) in the multi-round dialogue scenario, focus on usefulness, trustiness and reliability of the model responses, study of large-scale RAG techniques/biases in large-scale models.

PUBLICATION

Under review(first author, † corresponding author)***

Haoyi Zhang, **Shun Liu***, Siyuan Lin*, Zhaozhao Ma, Yang Han, Zhenghan Chen†, Jiechao Gao. "Adaptive Robust Classification: Integrating Distributional Resilience with Semi-supervised Learning in Medical Imagery", The 18th European Conference on Computer Vision (**ECCV'24**) **(CCF-B)***

*Weicheng Ma, Ethan Gearey, **Shun Liu**, Lili Wang, Soroush Vosoughi†. "Dissecting Values in Language Models: A Comprehensive Analysis of Stereotyping Behaviors and Their Implications in LLMs", (**ACL ARR 2024 February**) **(CCF-A)***

Shun Liu*,** Jianan Zhang, Ruocheng Song, Teik Toe Teoh†. "ADA-YOLO: Dynamic Fusion of YOLOv8 and Adaptive Heads for Precise Image Detection and Diagnosis", The Cognitive Science Society **(COGSCI'2024)** **(CCF-B)

In Progress

***Shun Liu*,** Jiaxi Yang†. "Model-Agnostic Interpretation Framework in Machine Learning: A Comparative Study in NBA Statistics"*

Shun Liu*†, *Kexin Wu, Chufeng Jiang, Danqing Ma. "Financial Time-Series Forecasting: Towards Synergizing Performances and Interpretability Within a Hybrid Machine Learning Approach"*

SKILL & HONOR

HONORS:

- **Gold** medal in China Collegiate Algorithm Design & Programming Challenge Contest
- **Top 1%** (Solo) Team in Kaggle CAFA 5 Protein Function Prediction
- **Top 2%** Team in Kaggle LLM Science Exam

SKILLS:

- **Technical Languages**: C++, Python, SQL, C#, HTML, CSS, JavaScript, Shell, Linux
- **Language**: English (Professional) and Mandarin (Native).
- **Tools**: MYSQL, Git, Github, LaTex
- **Interests**: Basketball, Reading